

Dynasties and Counterpunches (2000-2016)

Seventeen seasons of punch and counterpunch: Tampa 2 and the quarterback who solved it, a rule that tilted the field, the Patriots' shape-shifting offense, the 3-4's return, the read-option wave, and the first analysts.

February 2, 2014, Super Bowl XLVIII. The Denver Broncos arrive with the highest-scoring offense the NFL has ever seen: 606 points, 55 touchdown passes from Peyton Manning, five players with ten or more touchdowns. Seattle's defense lines up the way it has all season, one safety deep in the middle, cornerbacks pressed on the receivers' noses, an eighth defender walked up near the line, and does almost nothing fancy. Final score: Seattle 43, Denver 8. The broadcast keeps asking what Seattle did to Manning. The better question is older and bigger: how did the league get to a point where the greatest passing offense in history and a defense built to stop exactly that met in the final game? The answer is seventeen seasons of punch and counterpunch. A defense dominates, an offense finds the seam in it, the rules lean one way, a coach reinvents his team, the defense answers, and every answer becomes the next thing to beat. This chapter tells that story, from the Tampa 2 dynasty to the read-option wave, and shows you how to see its leftovers on any Sunday.

You'll be able to...

- Explain why Tampa 2 dominated the early 2000s and how Peyton Manning's Colts and the seam-running tight end beat it, and point to the throw on a diagram.
- Say what the 2004 illegal-contact emphasis changed and read the passing boom that followed in the league's efficiency numbers.
- Explain how the Patriots kept changing shape (an Erhardt-Perkins language, the 2007 spread, the 2010-2012 two-tight-end offense) and why a **move tight end** forces a defense to choose between base and nickel before it knows where he will line up.
- Explain the 3-4 revival of the 2000s and why the **hybrid edge** became one of the most valuable players on a defense.
- Describe the **2012 read-option wave**, what the data say about how big it really was, and why it receded; and place Seattle's Cover 3 and Chip Kelly's 2013 Eagles in the same story.
- Explain where football analytics came from (David Romer, Brian Burke, Football Outsiders) and the first game decision that put it on the front page.

i You'll need

This chapter continues the history of Part 11 and pulls together threads from all over the book. One line each, with a link if you want to catch up:

- **The college laboratory** (The College Laboratory): the spread offense, the spread option and the RPO were proved in college before the NFL took them. This chapter picks up as they cross over.
- **Tampa 2** (Cover 2 and Tampa 2): two deep safeties, five underneath, and a Mike linebacker who turns and runs down the middle to carry anything up the seam.
- **Personnel** (Personnel Groupings): two digits, backs then tight ends, so 12 personnel is one back and two tight ends. A defense answers with base (four defensive backs) or nickel (five).
- **The zone read** (Option Football): the quarterback leaves one defender unblocked and hands off or keeps depending on what that defender does.
- **The stand-up edge** (Odd and Hybrid Fronts): the 3-4's outside linebacker, a pass rusher who lines up standing at the end of the line.
- **EPA** (Downs, Distance, and Field Position): expected points added measures how much a play changed the offense's expected score. Positive is good for the offense.

Reading the diagrams: offense is blue circles (X, Y, Z, H receivers; F a second tight end; Q quarterback; R running back); defense is outlined squares (E and T ends and tackles; M the Mike, W the Will, S the Sam; \$ the nickel back; CB corners; FS and SS safeties). Solid arrows are routes and runs, dashed arrows defenders' movements, dotted lines man-coverage assignments, tees blocks. The dark blue line is the line of scrimmage and the orange line the line to gain.

The shape of the era

Start with the whole board. The figure below lays the seventeen seasons side by side: offensive moves on the left, defensive answers on the right, and in the middle the rule changes and new ideas that tilted the field. Read any row across and you see the rhythm of this chapter. Very little happened in isolation.

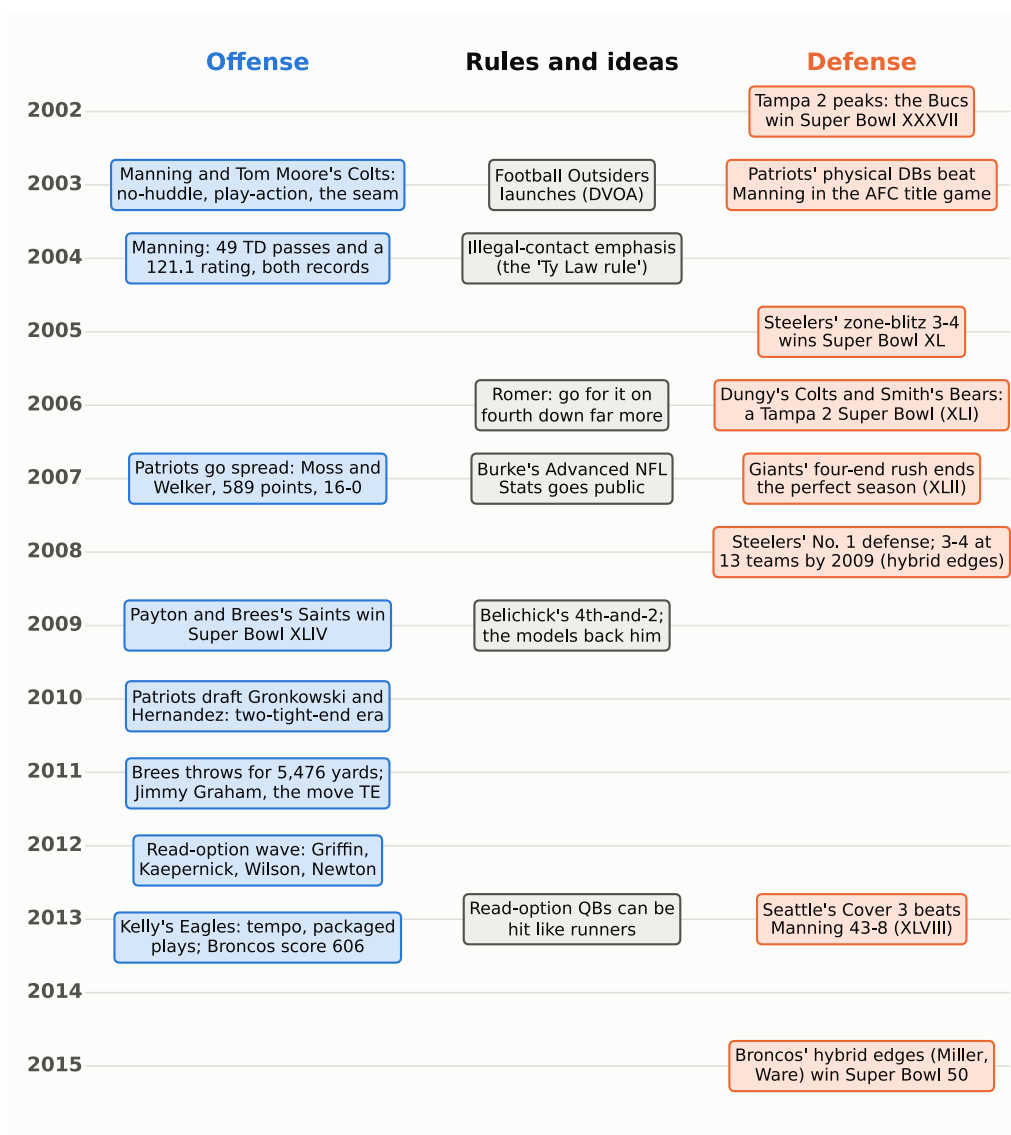


Figure 1: The era at a glance, 2002-2015. Offensive moves (blue, left), defensive answers (orange, right), and the rules and ideas in between (grey, centre). Each item is placed at its season; a Super Bowl is listed under the season it ended (Super Bowl XLVIII, played in February 2014, belongs to 2013). Read across a row to see a punch and the counterpunch.

Three forces push on every row.

- **Defenses built on speed and simplicity**, Tampa 2 above all, which made offenses earn every yard.
- **Rules that favoured the passing game**, starting with a crackdown on downfield contact in 2004 and continuing with new protections for quarterbacks and receivers.
- **Offenses that turned personnel into a weapon**: the no-huddle, the tight end who could be a receiver one snap and a blocker the next, and, from college, the quarterback who could run.

The defenses that won titles late in the era (the 2013 Seahawks, the 2015 Broncos) won by answering those forces with new kinds of players: press corners who could run, safeties who could hit, and pass rushers who were neither linemen nor linebackers. That is the arc. Now the details, in roughly the order they happened.

Tampa 2, and the quarterback who solved it

At the turn of the century the most copied defense in football was Tony Dungy and Monte Kiffin's **Tampa 2**: Cover 2 and Tampa 2 has its whole anatomy. In one paragraph: four fast linemen rush, two safeties guard the deep halves, two squat corners take the flats, and a rangy middle linebacker drops straight down the middle of the field at the snap to close the hole between the safeties. The defense keeps everything in front, tackles immediately, and waits for the offense to make a mistake. In 2002 Tampa Bay allowed the fewest points in the league and won Super Bowl XXXVII 48-21, intercepting the league MVP, Rich Gannon, five times.¹ By the middle of the decade Dungy's Colts, Lovie Smith's Bears and a string of imitators were running versions of it.

Why it won. Tampa 2 was a bet about the offenses of its day. In 2000 many NFL offenses still lined up with two backs and a tight end on early downs, threw short timing routes, and ran into a defense whose linebackers were faster than any blockers who could reach them. Against that, a defense of smaller, quicker players who all knew one simple scheme cold was terrifically hard to drive the ball against. Its weak spots, the deep middle behind the Mike, the sideline between corner and safety, the open middle at 12 to 15 yards when the Mike runs, were real, but hitting them took an accurate quarterback, patience, and a receiver who could beat a linebacker 20 yards downfield. In 2000 few offenses had all three.

Then Dungy moved to Indianapolis in 2002 and found the quarterback who did. From then on Peyton Manning practised against Dungy's Tampa 2 every day, and the Colts' offense under coordinator Tom Moore was built to beat exactly that kind of defense.² Three things made it work.

1. **A small playbook, run fast and changed at the line.** The Colts often went without a huddle, which kept the defense from substituting and from disguising much, and Manning chose or changed the play at the line of scrimmage after seeing the defense. A defense whose whole philosophy was *simple calls, played fast* found itself playing simple calls against a quarterback who knew them all.
2. **A run game that made the linebackers step up.** The Colts' bread-and-butter run was a stretch play (outside zone) to Edgerrin James, and their play-action passes were built to look exactly like it for the first second.
3. **A tight end who ran like a receiver.** Dallas Clark, a 2003 first-round pick, was not a blocker who could catch. He was a receiver in a tight end's body, and he was fast enough to make a linebacker running down the middle of the field look slow.

Put those together and you get the play that ate Tampa 2.

¹"Super Bowl XXXVII," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XXXVII (January 26, 2003: Buccaneers 48, Raiders 21; Rich Gannon, the 2002 league MVP, threw five interceptions, a Super Bowl record, three returned for touchdowns; Tampa Bay allowed the fewest points in the league in 2002, 12.3 per game).

²"Peyton Manning," Wikipedia: https://en.wikipedia.org/wiki/Peyton_Manning (Colts 1998-2011; Dungy head coach 2002-2008); "Tom Moore (American football coach, born 1938)," Wikipedia: [https://en.wikipedia.org/wiki/Tom_Moore_\(American_football_coach,_born_1938\)](https://en.wikipedia.org/wiki/Tom_Moore_(American_football_coach,_born_1938)) (Colts offensive coordinator 1998-2008, senior offensive coordinator from 2009); "Dallas Clark," Wikipedia: https://en.wikipedia.org/wiki/Dallas_Clark (first-round pick, 24th overall, 2003). On the Colts' no-huddle and Manning's line-of-scrimmage calls, see In-Game Adjustments and Tempo and its sources.

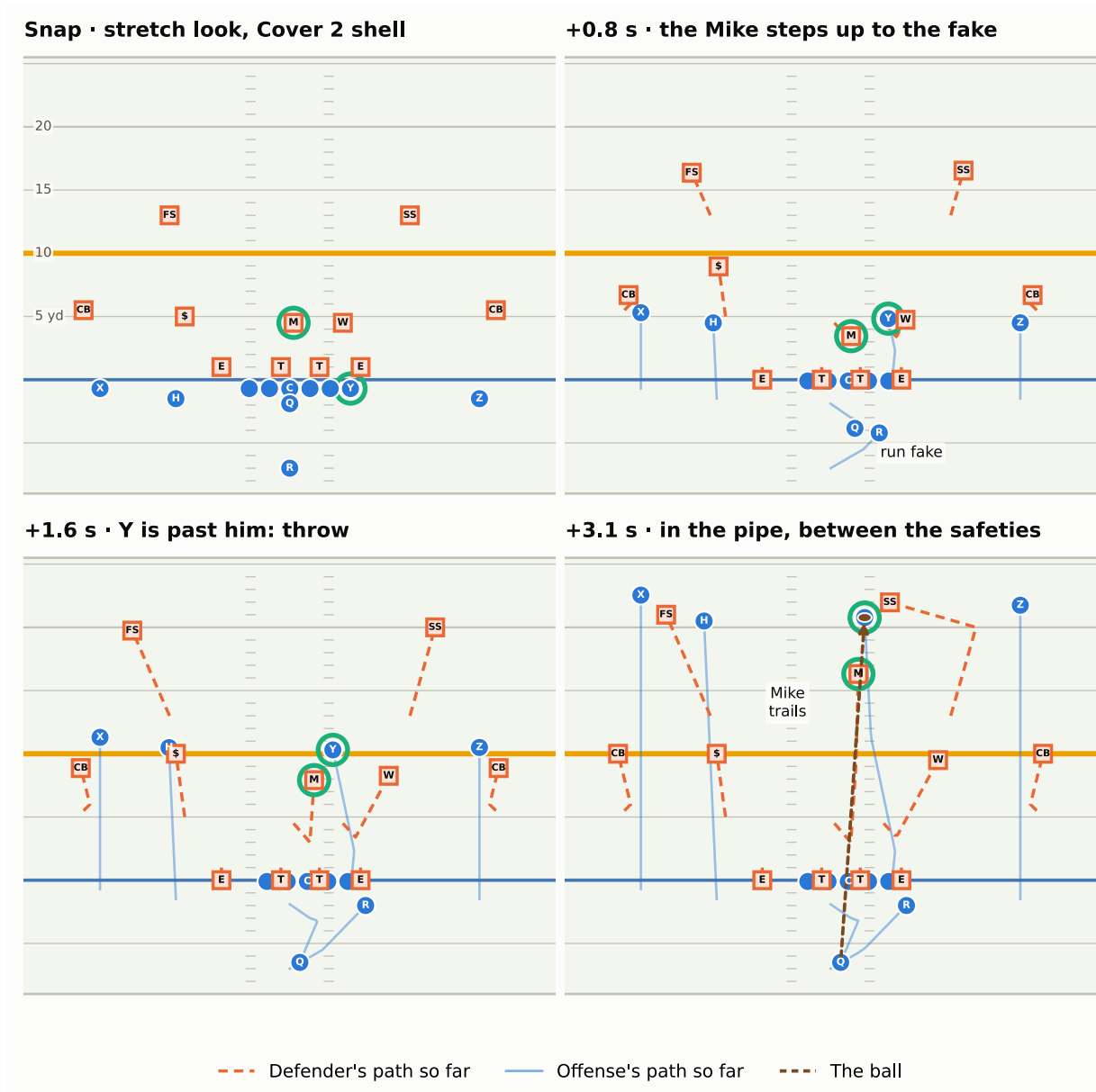


Figure 2: Manning against the Tampa 2 Mike, simulated (an illustration of the idea, not a specific Colts play). Snap: the Colts (11 personnel, under center) show their stretch run to the right; the defense shows a nickel Cover 2 with the Mike (M, ringed) 4.5 yards off the ball. +0.8 s: the Mike reads run and takes two steps up and toward the fake, as a linebacker must against the league's best run game. The tight end, Y (ringed), is already past him. +1.6 s: the Mike has turned to carry the seam, but he is three or four yards behind Y; the outside go routes have pulled both safeties wide, and Manning throws. +3.1 s: the ball arrives about 21 yards downfield in the pipe between the safeties, with the Mike four yards behind and the strong safety a step late. One false step by one defender is the whole play.

Follow the ringed Mike. Nothing he does is wrong. He reads his keys, sees the line step and the back run toward the sideline, and comes up to fill his gap, because a Tampa 2 Mike who ignores the run gets run over. By the time he sees the quarterback pull the ball back and turns to run, the tight end has a three- or four-yard head start, and a linebacker cannot make up three yards on a receiver over the next fifteen. Meanwhile the outside receivers' go routes have done their job: each safety has a wide receiver running at his half, so neither can cheat to the middle. The ball goes over the Mike and under the safeties.

That is the general shape of beating any defense, and it is worth stating once: **find the player with two jobs and make him do the wrong one first**. Tampa 2's Mike has a run job and a deep-pass job, and play-action puts them in the wrong order. Against a big, slow, blocking tight end the Mike could survive his false step. Against a receiver-speed tight end he couldn't. And if a defense answered by telling the Mike to bail at the snap, the Colts ran the stretch into a box one defender lighter. Manning's line-of-scrimmage checks were the last turn of the screw: he could see which answer the defense had chosen before the snap and pick the play that beat it.

Dungy's first trip back to Tampa as the Colts' head coach was a Monday night game against the defending champions and the defense he had built. For most of the night the Buccaneers' version of Tampa 2 won: an interception return by Ronde Barber made it 35-14 Tampa Bay with 5:09 left. Then the Colts scored three touchdowns in about four minutes (a Marvin Harrison catch among them), tied the game with 35 seconds left, and won 38-35 in overtime on a Mike Vanderjagt field goal.³

What to notice: how little the Buccaneers' defense changes when the Colts go no-huddle in the fourth quarter, and how Manning's completions move toward the middle of the field and the seams as the defense tires. A defense that wins by being simple and fast is in trouble when the offense is faster, and the drives that saved the game are a lesson in how to play the Tampa 2 patiently: short throws, quick snaps, then the shot over the middle when the linebackers' legs go.

Two seasons later the Colts' offense was the best in the league, and Manning's 2004 season (49 touchdown passes and a 121.1 passer rating, both records at the time) is still one of the great quarterback years.⁴ But it did not happen in a vacuum. Something else changed in 2004, and it changed the whole league.

2004: the rule that tilted the field

On January 18, 2004, in the AFC Championship Game, the New England Patriots beat the Colts 24-14 and intercepted Manning four times, three of them by cornerback Ty Law (safety Rodney Harrison had the other). New England's defensive backs played the Colts' receivers the way defensive backs had always played big games: they bumped, held and rerouted them well past the line of scrimmage, and the officials let them. Colts president Bill Polian, a member of the league's competition committee, complained loudly, and for the 2004 season the league made illegal contact a **point of emphasis**: an instruction to officials to enforce an existing rule more strictly. The rule itself was old. Since 1978 a defender may contact a receiver only within 5 yards of the line of scrimmage; beyond that, contact that impedes him before the ball is thrown is illegal contact, a 5-yard penalty with an automatic first down. What changed was that officials started throwing the flag: by STATS's count, illegal-contact calls went from 76 in the 2003 season to 190 in 2004. Patriots fans still call it the "Ty Law rule."⁵

Why it mattered so much. A flag that was rarely called is, in practice, not a rule. For decades defenders had been able to *shape* routes with their hands: knock a receiver off his stem at 7 yards, lean on him at 10, give him a shove as he broke. Every Cover 2 and Tampa 2 depended on some of that, because the coverage needs receivers to arrive late and in the wrong place. Take the hands away beyond 5 yards and every route arrives on time, and a timing offense like Manning's lives on routes that arrive on time. The change also made **man coverage** harder: a defender who can't use his hands downfield has to run with a receiver, not wrestle him.

Here is what the league's passing game did next.

³Pro Football Reference box score, Indianapolis Colts at Tampa Bay Buccaneers, October 6, 2003: <https://www.pro-football-reference.com/boxscores/200310060tam.htm> (Ronde Barber's 29-yard interception return made it 35-14 at 5:09 of the fourth quarter; a Ricky Williams 1-yard run tied it with 35 seconds left; Mike Vanderjagt's 29-yard field goal won it in overtime); Colts.com, "Looking back: October 6, 2003": <https://www.colts.com/news/looking-back-october-6-2003-5939404> (trailed 35-14 with four minutes remaining; 21-point comeback).

⁴"Peyton Manning," Wikipedia: https://en.wikipedia.org/wiki/Peyton_Manning (2004: 49 touchdown passes, breaking Dan Marino's 1984 record of 48; 121.1 passer rating, then the single-season record). Both records have since been broken.

⁵Tom E. Curran / NBC Sports Boston, "They called it the 'Ty Law Rule' ... and that was OK with him" (Aug. 3, 2019): <https://www.nbcsportsboston.com/nfl/new-england-patriots/they-called-it-the-ty-law-rule-and-that-was-ok-with-him/393549/> (Polian's complaints after the January 2004 AFC Championship Game, Patriots 24, Colts 14, in which Law intercepted Manning three times; the 2004 emphasis on downfield contact). Manning's four interceptions (three by Law, one by Rodney Harrison in the end zone): "2003-04 NFL playoffs," Wikipedia: https://en.wikipedia.org/wiki/2003%E2%80%9404_NFL_playoffs. Foul counts: Associated Press, "Rise in flags makes NFL defenders wary of contact" (Aug. 2014): <https://www.washingtoninformer.com/rise-in-flags-makes-nfl-defenders-wary-of-contact/> ("Illegal contact calls went from 76 the previous season to 190, according to STATS"). A widely repeated alternative count, 79 to 191, comes from Mike Florio, ProFootballTalk, "New point of emphasis results in plenty of flags for illegal contact, defensive holding" (Aug. 8, 2014): <https://www.nbcsports.com/nfl/profootballtalk/rumor-mill/news/new-point-of-emphasis-results-in-plenty-of-flags-for-illegal-contact-defensive-holding> ("The fouls climbed from 79 in the entire 2003 regular season to 191 in 2004"), whose wording may include defensive holding. The 5-yard contact limit dates from 1978 (1978 and the Passing Revolution); illegal contact today: NFL Rulebook, Rule 8, Section 4, <https://operations.nfl.com/the-rules/nfl-rulebook/>.

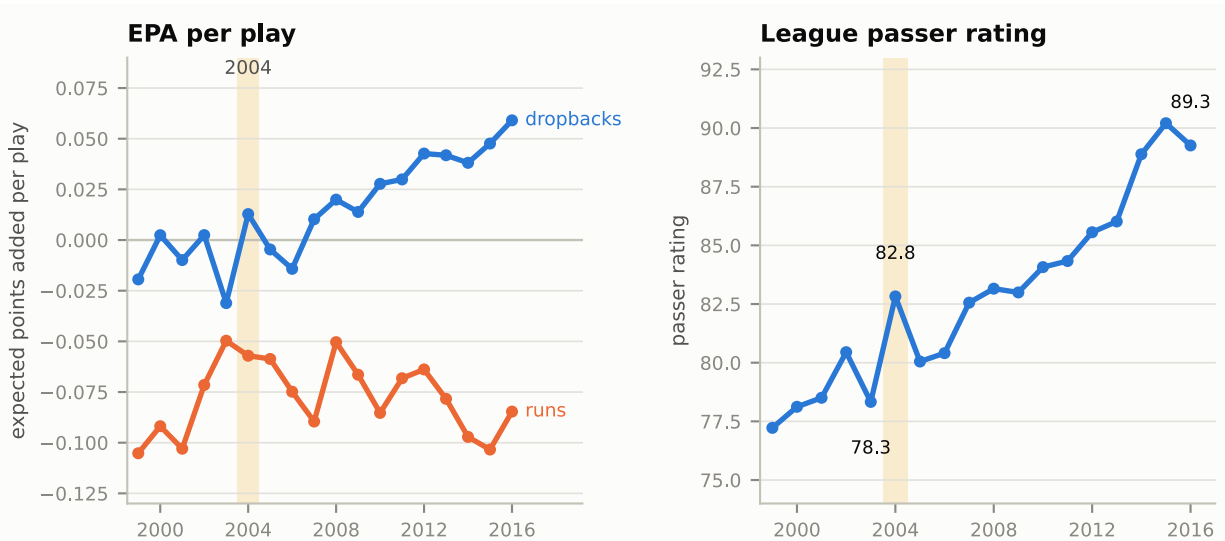


Figure 3: The passing boom, 1999-2016. Left: expected points added (EPA) per play, for dropbacks (passes, sacks and scrambles) and for designed runs, league-wide in the regular season. Throwing was already worth more than running in 1999; after 2004 the gap widened, as dropbacks became steadily more valuable and runs did not. Right: the league’s passer rating, the familiar box-score measure, which jumped by more than four points in 2004, the first season of the illegal-contact emphasis (shaded), and climbed about twelve points over the era. The dip in 2005 is a reminder that one season is noisy; the trend is the story. Source: nflverse play-by-play.

The pattern is unmistakable, and it has three parts.⁶

- **2004 was a jump.** League passer rating rose from about 78 to about 83 in a single season, and EPA per dropback went from negative to positive. Not all of that is the rule: Manning had a historic year, and seasons bounce. But the timing is hard to ignore, and the officials’ foul counts show the rule really did change what defenders could get away with.
- **The rise kept going.** Every rule change of the next decade pointed the same way: a 2009 directive that a defender knocked to the ground may not lunge at a quarterback’s legs (issued after Tom Brady’s knee was wrecked by such a hit in the 2008 opener), a 2009 rule against hits to the head of a “defenseless” receiver and wider protections after it, stricter roughing-the-passer enforcement.⁷ By 2016 the league passer rating was near 89, and a quarterback who would have been average in 2000 was below average.
- **Running didn’t keep up.** EPA per designed run barely moved. The gap between a dropback and a run grew by about two-thirds across the era, and the share of plays that were passes crept up with it. Every coach in the next section was making the same calculation: the field had tilted toward the passing game, so the question was how to throw more, and better, without getting your quarterback hurt.

🔥 Common misconception

“The NFL changed the rules in 2004.” Not quite, and the difference matters if you want to understand how the league works. The 5-yard contact limit had been in the rulebook since 1978. In 2004 the league told officials to *call* it. Most of the NFL’s big strategic shifts work that way: a “point of emphasis” or a new interpretation changes the game as much as a new rule does, and it can quietly fade a few seasons later. When a broadcaster says “they’re calling it tighter this year,” take it seriously.

⁶This book’s calculation from nflverse play-by-play, regular seasons 1999-2016. Dropbacks are plays nflverse flags `qb_dropback` (passes, sacks and scrambles); designed runs are the other run plays. EPA per dropback: -0.031 (2003), $+0.013$ (2004), $+0.059$ (2016); per designed run: -0.050 (2003), -0.057 (2004), -0.085 (2016). League passer rating from the standard formula on pass attempts (no sacks or two-point tries): 78.3 (2003), 82.8 (2004), 80.0 (2005), 89.3 (2016). The pass share of plays rose from 55% (2003-2006) to 60% (2015-2016). The nflverse expected-points model is fitted across all seasons, so league-average EPA can drift for reasons other than play quality; the passer-rating panel is a check that does not depend on the model. Reproduce with the code behind the figure.

⁷NFL.com / Associated Press, “Owners pass four rules in an attempt to make the game safer” (March 2009): <https://www.nfl.com/news/owners-pass-four-rules-in-an-attempt-to-make-the-game-safer-09000d5d80f6c090> (adopted for 2009: “Initial contact to the head of a defenseless receiver also will draw a 15-yard penalty”; separately, as an officiating adjustment that needed no owners’ vote, “Defenders who are knocked to the ground no longer can lunge into quarterbacks,” after Bernard Pollard’s hit ended Brady’s 2008 season in the opener; often called the “Brady rule”). A 2006 rule (the “Carson Palmer rule”) had already barred most hits at or below a passer’s knees. On later defenseless-player protections, see Contact Rules and Penalties as Strategy.

There is an irony in the story, and it is worth saying aloud. The Colts' complaints changed the rules, and Dungy, whose Tampa 2 depended on physical underneath coverage, was the Colts' head coach. The change helped his quarterback and hurt his scheme, and the league's Tampa 2 copies started fading at about the same time. The first team to show the league what the new passing game could do, though, was not the Colts. It was the team that had beaten them.

New England: the team that kept changing shape

Bill Belichick's Patriots won three Super Bowls in four seasons (XXXVI, XXXVIII and XXXIX, after the 2001, 2003 and 2004 seasons) with a different team almost every year, and what stayed constant was not a scheme but a habit: decide what this opponent, this season, can't handle, and become that. The full case study is Belichick and Game-Plan Football. Here are the three shapes that mattered to the league's arms race.

The language underneath. The Patriots' offense spoke Erhardt-Perkins, the concept-based play-calling system (Offensive Systems and Languages has it): one word names a whole pass pattern, and the same word works from any formation. That is what made the next two shapes possible. A team that can run its concepts from any formation and any personnel group can change both every week without teaching a new playbook.

2007: the spread. Before the 2007 season New England traded for two receivers, Randy Moss (from Oakland, for a fourth-round pick) and Wes Welker (from Miami), and became something the NFL had not seen from a contender: a spread offense, often from the shotgun with three and four receivers, Moss running past people outside and Welker running option routes from the slot. The Patriots scored 589 points, a record at the time; Brady threw 50 touchdown passes and Moss caught 23, both records; they won all sixteen regular-season games.⁸ The 2007 Patriots were the NFL's first great answer to the 2004 rules: spread the field, put the ball in the air, and make the defense cover all 53 yards of width with defenders who couldn't touch anyone past 5 yards.

The counterpunch: four men who could get there. They lost the Super Bowl. In Super Bowl XLII, on February 3, 2008, the New York Giants beat them 17-14, and they did it with the oldest answer to a spread passing game: a four-man pass rush good enough to get home without blitzing. Defensive coordinator Steve Spagnuolo's Giants had a surplus of defensive ends (Michael Strahan, Osi Umenyiora and Justin Tuck; Mathias Kiwanuka, an end moved to linebacker that year, had broken his leg in November), and on passing downs he put three or even four ends on the field at once, with Tuck sliding inside over a guard. (Four years later, when the Giants beat New England in another Super Bowl, their players had a name for that speed package: "NASCAR.") They sacked Brady five times and hit him again and again.⁹ The arithmetic is the whole lesson: if four rushers win, seven defenders can cover the five receivers a spread puts out, with two left over to help. Spagnuolo, more than eighteen years later, is still coordinating a defense on that idea, in Kansas City.

2010-2012: two tight ends, and the move tight end

Three years later the Patriots changed shape again, in almost the opposite direction. In the 2010 draft they took two tight ends: Rob Gronkowski in the second round and Aaron Hernandez in the fourth. By 2011 they were the core of the offense. Gronkowski was the most complete tight end in football, a 6-foot-6, 265-pound blocker who caught 17 touchdown passes and gained 1,327 receiving yards that season, both records for a tight end.¹⁰ Hernandez was something else: a tight end in name who lined up in the slot, out wide, next to the tackle, and in the backfield as a running back. In a playoff game against Denver in January 2012 he took a handoff on the Patriots' first drive and ran 42 yards.¹¹

⁸"2007 New England Patriots season," Wikipedia: https://en.wikipedia.org/wiki/2007_New_England_Patriots_season (589 points, then the NFL record; 16-0 regular season; Brady's 50 touchdown passes and Moss's 23 touchdown catches, both records at the time; Moss acquired from Oakland for a 2007 fourth-round pick, No. 110, which New England had obtained from San Francisco; Welker from Miami for the Patriots' 2007 second- and seventh-round picks). The 2013 Broncos' 606 points later broke the scoring record.

⁹"Super Bowl XLII," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XLII (February 3, 2008, Glendale, Arizona; Giants 17, Patriots 14; Brady sacked five times). Spagnuolo "often used a smaller defensive line, with three or even four defensive ends": "Steve Spagnuolo," Wikipedia: https://en.wikipedia.org/wiki/Steve_Spagnuolo; Strahan and Umenyiora outside and Tuck inside over a guard: NFL.com, "Spagnuolo's schemes worked to perfection" (Feb. 2008): <https://nfl.com/news/spagnuolo-s-schemes-worked-to-perfection-09000d5d8067edb1>. Kiwanuka's move to strongside linebacker and his broken fibula (Nov. 18, 2007, season-ending): "Mathias Kiwanuka," Wikipedia: https://en.wikipedia.org/wiki/Mathias_Kiwanuka. The "NASCAR" name, in Justin Tuck's words at the time of Super Bowl XLVI ("We came up with NASCAR; we call it our speed package"): Jayski, "NASCAR propels Giants to Super Bowl victory" (Feb. 7, 2012): <https://www.jayski.com/2012/02/07/nascar-propels-giants-to-super-bowl-victory/>; an SI profile calls the 2007 version the "Four Aces" package: <https://www.si.com/nfl/2016/12/27/nfl-new-york-giants-steve-spagnuolo-defensive-coordinator>. Spagnuolo remains the Chiefs' defensive coordinator in 2026 (course fact sheet, 2026 staffs).

¹⁰"Rob Gronkowski," Wikipedia: https://en.wikipedia.org/wiki/Rob_Gronkowski (2010 draft, second round, 42nd overall; 2011: 90 receptions, 1,327 yards and 17 receiving touchdowns, the tight-end records for touchdowns and yards at the time).

¹¹"Aaron Hernandez," Wikipedia: https://en.wikipedia.org/wiki/Aaron_Hernandez (2010 draft, fourth round, 113th overall; 2011: 79 receptions, 910 yards; used as an H-back and in the slot; a 42-yard carry on the Patriots' first drive of the January 14, 2012 divisional playoff against Denver). His career ended when he was charged with murder in June 2013 and released by the Patriots the same day.

Hernandez was the archetype of a player the whole league would soon want, the **move tight end**, also called the **joker**: a tight end who doesn't have a fixed spot, so the defense can't plan for where he'll be. (The diagrams label him F; many staffs call the move tight end the F or the H, and the in-line tight end the Y.) The point of him is not any one route. It is the decision he forces on the defense before the snap.

Recall the rule of substitution from Personnel Groupings: the offense sends its players on first, the defense answers with its own, and only then does the offense pick a formation. With 12 personnel on the field, a defense normally chooses between two answers, and the move tight end makes both of them wrong.

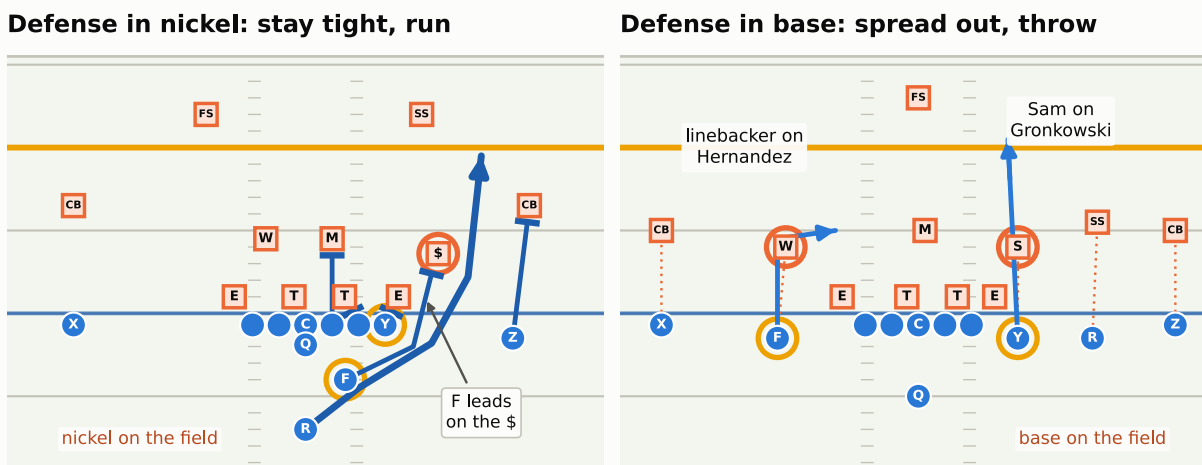


Figure 4: The Patriots' 12-personnel dilemma, 2011 style (illustrative). The same eleven players, Gronkowski (Y) and Hernandez (F) among them (gold rings), after the defense has made its personnel choice. Left: the defense matched two tight ends with nickel, so the Patriots line up tight with Hernandez in the backfield as a lead blocker: a 250-pound tight end leads into the 195-pound nickel back (\$), Gronkowski drives the end, and the back follows them outside. Right: the defense matched with its base 4-3, so the Patriots spread to an empty backfield: Hernandez in the left slot draws the Will linebacker (W) in man coverage, Gronkowski is detached on the right with the Sam (S) on him, and the running back is split wide against the strong safety. Dotted lines are man assignments. Either way, the offense gets the matchup it wanted, because it chose the formation after the defense chose its players.

Write the dilemma out as a table and you can see why defensive coordinators hated it. The defense chooses a row before the snap; the offense, seeing the choice, chooses a column.

Defense sends out...	Offense lines up tight (run)	Offense spreads out (pass)
Base (3 linebackers, 4 DBs)	Fine for the defense: big bodies against a run	Bad: linebackers in man coverage on two receiver-speed tight ends
Nickel (2 linebackers, 5 DBs)	Bad: a 195-pound nickel back is the run's target, and the tight ends block him	Fine for the defense: coverage players against a spread

The offense just takes whichever bad cell the defense handed it. Ordinary two-tight-end teams couldn't do this, because their tight ends were good at only one of the two jobs: a blocking tight end split out wide is no threat, and a receiving tight end in-line can't seal a defensive end. The Patriots' pair could do both, and Hernandez could do both from anywhere. Then New England added the last piece: tempo. A **no-huddle** offense that snaps the ball before the defense can substitute keeps the defense in whichever personnel it had on the last play. Find the mismatch once, then snap it again before they can fix it. In 2012 the Patriots ran 1,191 offensive plays, one of the highest totals in NFL history, and scored 557 points.¹²

¹²Sports Illustrated scouting reports, Sept. 2, 2013: <https://vault.si.com/vault/2013/09/02/scouting-reports> (1,191 plays in 2012, the second-most in NFL history); "2012 New England Patriots season," Wikipedia: https://en.wikipedia.org/wiki/2012_New_England_Patriots_season (557 points; a record 444 first downs).

! Why it exists: the move tight end

What it solves. Defenses decide their personnel *before* the formation. A player who is a credible blocker and a credible receiver, and who can line up anywhere, turns that order of events into a guaranteed mismatch: whatever the defense picks, he goes where it is weakest.

What it costs. He is rare: the body to block a defensive end and the speed to beat a safety rarely come together, and the offense is stuck when he gets hurt. A move tight end who is really a big receiver (a poor blocker) gives the defense its answer back: play nickel, and dare the offense to run. And he puts the tight end's salary in a strange place, as the next story shows.

New Orleans had one too. Sean Payton's Saints, with Drew Brees, won Super Bowl XLIV after the 2009 season and in 2011 set the record for yards in a season while Brees threw for 5,476 yards, then a record.¹³ Their offense was built on the same principle from a different direction: many personnel groups, many formations, and players the defense couldn't label. The cleanest example is Jimmy Graham, a former college basketball player who became a 6-foot-7 tight end. In 2014, Graham and the players' union argued that he should be paid under the franchise tag as a *wide receiver* (about \$12 million) rather than a tight end (about \$7 million), because in 2013 he had lined up in the slot or out wide on about two-thirds of his snaps. An arbitrator ruled he was a tight end.¹⁴ The argument itself is the history lesson: by the early 2010s the tight end had become a position you could no longer define by where he stood.

i Film room: Patriots vs. Broncos, divisional playoff, January 14, 2012

New England beat Denver 45-10, and Brady threw six touchdown passes.¹⁵ The game is a showcase for the 2011 offense.

What to notice: the substitutions, or rather the lack of them. Watch Denver's sideline as the Patriots go no-huddle with both tight ends on the field, and count how often the defense is still in the personnel it had for the previous play. Then watch Hernandez: in the backfield on the first drive (his long run), in the slot on later snaps, in-line on others. Every time he moves, ask yourself who is covering him and whether that player was chosen for him or stuck with him.

The 3-4 revival and the hybrid edge

While offenses were spreading out, defenses were changing shape too, and their biggest change went the other way from Tampa 2's small, fast linemen. At the start of the decade the 3-4 (three defensive linemen, four linebackers) was a rarity: by Bill Belichick's later count, only two teams, the Steelers and his own Patriots, ran it in 2000. By 2009 thirteen of the league's 32 teams lined up in some form of it.¹⁶ The story of how it got there, and the two-gap and one-gap branches, is in Odd and Hybrid Fronts. This chapter cares about *why* in the 2000s, and the answer is a kind of player.

Pittsburgh showed the way. Dick LeBeau's Steelers defense won Super Bowl XL after the 2005 season and Super Bowl XLIII after 2008, when it allowed the fewest points and yards in the league and outside linebacker James Harrison, an undrafted free agent,

¹³"New Orleans Saints," Wikipedia: https://en.wikipedia.org/wiki/New_Orleans_Saints (Super Bowl XLIV, February 7, 2010: Saints 31, Colts 17); "2011 New Orleans Saints season," Wikipedia: https://en.wikipedia.org/wiki/2011_New_Orleans_Saints_season (7,474 offensive yards, an NFL single-season record, breaking the 2000 Rams' 7,075; Brees's 5,476 passing yards, breaking Dan Marino's 5,084); Guinness World Records, "Most offensive yards in an NFL season (team)": <https://www.guinnessworldrecords.com/world-records/102033-most-offensive-yards-in-an-nfl-season-team>. Manning broke Brees's passing-yardage record by one yard in 2013.

¹⁴Sports Illustrated, "Arbitrator rules Saints' Jimmy Graham is a tight end for franchise tag purposes" (July 2014): <https://www.si.com/nfl/2014/07/01/arbitrator-ruling-saints-jimmy-graham-franchise-tag-grievance>; ABC News/ESPN, "Jimmy Graham ruled a tight end": <https://abc7.com/post/jimmy-graham-ruled-a-tight-end/151701/> (arbitrator Stephen Burbank's July 2, 2014 ruling; Graham lined up in the slot or out wide on 67 percent of his 2013 snaps; tight-end tender \$7.053 million against about \$12.1 million for a receiver). Graham appealed, then signed a long-term deal with the Saints before the appeal was heard.

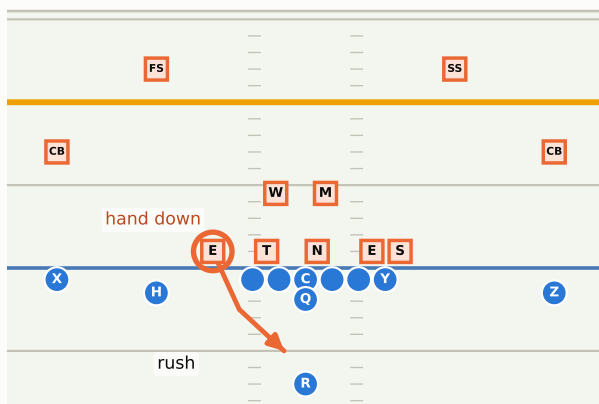
¹⁵"2011 New England Patriots season," Wikipedia: https://en.wikipedia.org/wiki/2011_New_England_Patriots_season (AFC Divisional playoff, January 14, 2012: Patriots 45, Broncos 10; Brady "tied an NFL playoff record by throwing six touchdown passes," five of them in the first half).

¹⁶NFL.com, "Teams incorporating 3-4 defense to add pressure on quarterbacks" (2009): <https://amp.nfl.com/news/teams-incorporating-3-4-defense-to-add-pressure-on-quarterbacks-09000d5d8124eb61> (with Denver, Kansas City and Green Bay switching in 2009, "13 of the 32 teams now line up in some form of the 3-4"). For 2000: Jeremy Fowler, ESPN, "Bill Belichick recalls days when Patriots and Steelers set 3-4 defense trend" (April 9, 2016): https://www.espn.com/blog/pittsburgh-steelers/post/_/id/18346/bill-belichick-recalls-days-when-patriots-and-steelers-set-3-4-defense-trend (Belichick "reached back to 2000, when two NFL teams ran a 3-4 defense," the Patriots and Steelers). Wikipedia's "3-4 defense" article says the Steelers were the only 3-4 team in 2001, and Belichick in the same talk said half the league played a 3-4 by 2005, more than NFL.com's 2009 count. Counts differ by source, depending on whether a team's base defense or any 3-4 use is counted.

was Defensive Player of the Year.¹⁷ LeBeau's zone blitzes (Zone Blitz, Fire Zones, and Simulated Pressure) depended on players at the end of the line who could either rush or drop into coverage, so the offense never knew which four or five were coming. New England built its 3-4 the other way, around two-gap linemen and versatile linebackers like Mike Vrabel (who, a sign of the times, also caught touchdown passes as a goal-line tight end, two of them in Super Bowls).¹⁸ Wade Phillips's 3-4 in Dallas made DeMarcus Ware a 20-sack rusher in 2008. Green Bay switched to Dom Capers's 3-4 in 2009 and drafted Clay Matthews to stand at the end of it.

What these defenses had in common is the player the 3-4 is built to use: the **hybrid edge**. He is part defensive end and part outside linebacker: big enough to set the edge against a 310-pound tackle, fast enough to rush from a standing start, and agile enough to drop into a short zone. A 4-3 team lists him as a defensive end and a 3-4 team as an outside linebacker, and on the field he stands in nearly the same place.

4-3 Under: a defensive end



3-4: an outside linebacker

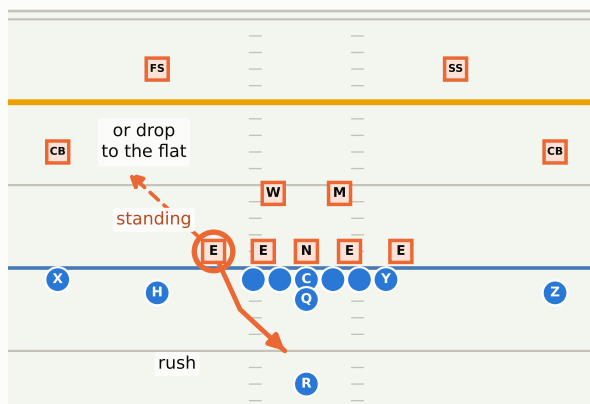


Figure 5: One player, two labels. The same 11-personnel offense against a 4-3 Under front (left) and a 3-4 (right). The ringed player on the defense's right (the offense's left) is the same kind of player in both: in the 4-3 he is the weak-side defensive end with a hand on the ground, in the 3-4 the weak-side outside linebacker, standing. He lines up in almost the same spot, a yard or so outside the left tackle. In the 4-3 his menu is mostly one item, rush (solid arrow); in the 3-4 the offense also has to respect a drop into the flat (dashed arrow), and it can't tell which until the snap. That uncertainty is what the 3-4 of the 2000s was selling, and the hybrid edge is what made it work.

Why it exists. Three reasons, each tied to the rest of this chapter.

1. **Body types.** College football in the 2000s produced a flood of 240- to 260-pound pass rushers, too light to hold up as a 4-3 defensive end against the run on every down, too explosive to waste at linebacker. The 3-4 gave them a job: stand at the end of the line, rush most of the time, and let the big three-man line in front of the linebackers do the heavy work.
2. **The passing boom.** After 2004 the quarterback had more time and cleaner routes; the defense's best counter was to hit him sooner. A front where any of five or six players might rush, and where one who looks like a rusher might drop, makes protection harder to set (Pass Protection explains why a protection needs to know who is coming).
3. **Disguise against spread offenses.** A 4-3 end with his hand down is a rusher; everybody knows it. A standing edge can be a rusher or a dropper, and an offense that spreads five receivers out can't afford a guess.

What it costs. The 3-4 needs a nose tackle and two ends who can occupy blockers, and those 300-plus-pound players are rare and expensive. The hybrid edge himself is a compromise: lighter than a defensive end, he can get pushed around against power runs. And the more teams wanted these players, the more they cost, which is the real legacy of the era: by the 2010s draft boards and analysts were listing "EDGE" as a position of its own, regardless of the front a team played, and the best of them became the

¹⁷"2008 Pittsburgh Steelers season," Wikipedia: https://en.wikipedia.org/wiki/2008_Pittsburgh_Steelers_season (first in the league in points and yards allowed; Super Bowl XLIII); "James Harrison (American football)," Wikipedia: [https://en.wikipedia.org/wiki/James_Harrison_\(American_football\)](https://en.wikipedia.org/wiki/James_Harrison_(American_football)) (undrafted out of Kent State in 2002; 2008 AP Defensive Player of the Year); "Super Bowl XL," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XL (February 5, 2006: Steelers 21, Seahawks 10). LeBeau was Pittsburgh's defensive coordinator 2004-2014: https://en.wikipedia.org/wiki/Dick_LeBeau. DeMarcus Ware's 20 sacks in 2008: https://en.wikipedia.org/wiki/DeMarcus_Ware; Green Bay's 2009 switch to Capers's 3-4 and drafting of Clay Matthews: https://en.wikipedia.org/wiki/Clay_Matthews_III; Von Miller, second overall in 2011: https://en.wikipedia.org/wiki/Von_Miller; the 2015 Broncos and Super Bowl 50: Odd and Hybrid Fronts and its sources.

¹⁸"Mike Vrabel," Wikipedia: https://en.wikipedia.org/wiki/Mike_Vrabel (Patriots linebacker 2001-2008; touchdown catches as an eligible lineman/tight end on goal-line plays, a 1-yard catch in Super Bowl XXXVIII and a 2-yard catch in Super Bowl XXXIX; the article's body counts 12 career receptions, all touchdowns, though its infobox says 10). Vrabel was the Patriots' head coach in 2025 and the AP Coach of the Year (course fact sheet).

highest-paid defenders in the game. Von Miller, the second pick in 2011, and DeMarcus Ware led the 2015 Broncos, who allowed the fewest yards in the league and won Super Bowl 50 by hitting the league MVP, Cam Newton, again and again. That game's film room is in Odd and Hybrid Fronts, and the edge's rise to the most valuable defender on the field is the subject of The Modern Era.

i Madden vs. real life

In a video game, "3-4" and "4-3" are menus you pick before the play, and the positions on the depth chart decide who plays where. On a real field the labels are close to marketing. A 3-4 team in nickel (which is most of the time on passing downs) usually lines up with four players at the line, two of them standing hybrid edges, and looks a lot like a 4-3 team in nickel. What tells you the defense's philosophy is not the label but what its edges do: hand down and rush every snap, or stand up and make the quarterback wonder.

Seattle's answer: be simple, be bigger

By 2012 the offense had the momentum: spread formations, receiver-speed tight ends, rules that kept defenders' hands off. Pete Carroll's Seattle Seahawks answered by going back to basics with a new kind of player. Their defense played mostly one coverage, Cover 3 with a single deep safety, and dared the offense to beat it, but filled it with unusually big, long press cornerbacks (Richard Sherman was 6-foot-3), a free safety with a sprinter's range (Earl Thomas), and a strong safety built like a linebacker (Kam Chancellor). The press corners used every inch of the 5 yards the rules still allowed, and the big safety let the defense put an eighth player near the line against the run without bringing in a linebacker who couldn't cover. Seattle allowed the fewest points in the league four seasons running, and in Super Bowl XLVIII it held Manning's record-setting Broncos to 8 points.¹⁹

Notice how neatly Seattle answered the rest of this chapter. Against the move tight end, a strong safety who could tackle like a linebacker meant the defense didn't have to choose between base and nickel. Against the post-2004 passing game, corners who could jam *within* the 5-yard zone and then run with receivers made the rule matter less. Against tempo, one coverage played perfectly is hard to catch out of position. The full anatomy, the match rules that carried seams and the coaches who spread it, is in The Cover 3 Family.

The 2012 read-option wave, and why it receded

Every innovation in this chapter so far came from NFL coaches reacting to NFL problems. The next one came from college. By the late 2000s the spread option (The College Laboratory) dominated college football, and its quarterbacks were arriving in the NFL. In 2012 four of them ran the zone read for NFL teams at once: Robert Griffin III, a rookie in Washington's pistol offense; Colin Kaepernick, who took over San Francisco's offense in midseason; Russell Wilson, a rookie in Seattle; and Cam Newton, in his second season in Carolina. That burst is remembered as the **read-option wave**: a season in which it looked as if NFL offenses were about to turn into college ones. Option Football tells it play by play, including Kaepernick's record 181 rushing yards against Green Bay in the playoffs.

Why it worked. The zone read is an arithmetic trick: by leaving one defender unblocked and *reading* him instead, the offense gets an extra blocker somewhere else, and a quarterback who is a real running threat makes that defender wrong whatever he does. NFL defenses in 2012 had spent a decade preparing for pocket passers and had little practice against it. And as the chart below shows, it was efficient: designed quarterback runs that year were worth about a tenth of a point per play, while ordinary running back runs lost value.

But how big was the wave, really? The data give an answer that surprises most people who remember it.

¹⁹"Super Bowl XLVIII," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_XLVIII (February 2, 2014, MetLife Stadium; Seahawks 43, Broncos 8); "2013 Denver Broncos season," Wikipedia: https://en.wikipedia.org/wiki/2013_Denver_Broncos_season (606 points, the NFL record; Manning's 55 touchdown passes and 5,477 yards, both records; five players with 10 or more touchdowns). Seattle led the league in fewest points allowed in 2012, 2013, 2014 and 2015, the first team to do it four years running since the 1953-57 Browns: Seahawks.com, "Seahawks 'incredibly proud' to lead NFL in scoring defense for fourth straight year" (Jan. 2016): <https://seahawks.com/news/seahawks-incredibly-proud-to-lead-nfl-in-scoring-defense-for-fourth-strai-147146>. Its personnel: The Cover 3 Family and its sources.

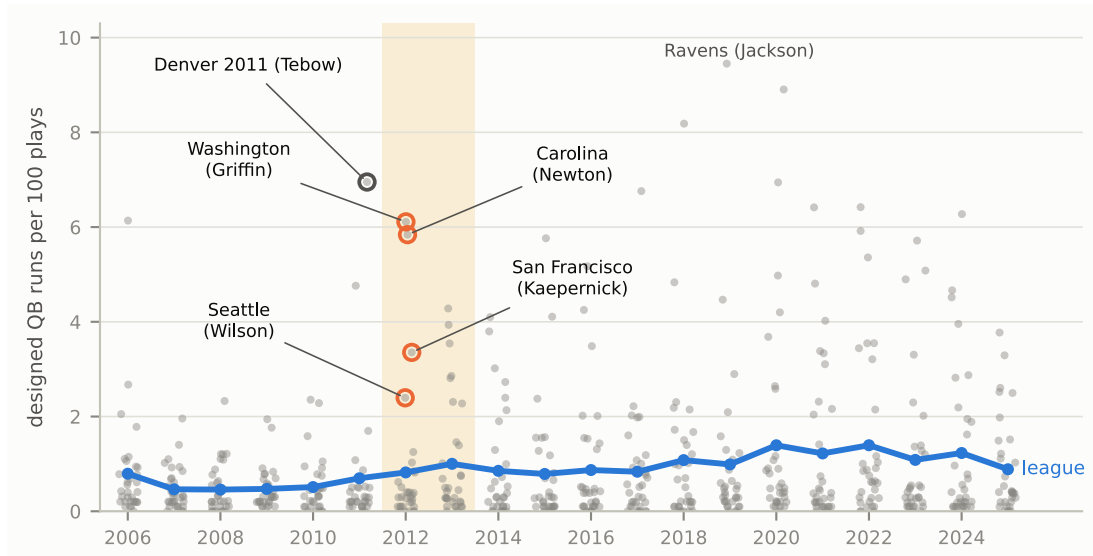


Figure 6: Designed quarterback runs per 100 offensive plays, 2006-2025: each grey dot is one team-season, the line is the league rate. The 2012 wave (shaded) is easy to see in a few teams (Washington with Griffin, Carolina with Newton, San Francisco with Kaepernick, Seattle with Wilson, ringed in orange; Tim Tebow's 2011 Broncos, the year before, ringed in grey) and hard to see in the league line, which crept up from about half a run per 100 plays to about one and then flattened. Those four teams ran about two-thirds of the league's designed quarterback runs in 2012. The bigger, league-wide rise came later, from 2018, led by Lamar Jackson's Ravens (the dots above 8). A designed run here is a non-scramble rush by a team's quarterback on 3 or more yards to go, so it leaves out sneaks; a read-option play the quarterback hands off is counted as a running back run, so this measures keeps, not reads. Source: nflverse play-by-play, regular seasons.

The shaded seasons tell the story in two parts. A handful of teams really did change: Washington and Carolina ran designed quarterback runs at a rate only two teams had reached in the previous six seasons (Michael Vick's 2006 Falcons and Tim Tebow's 2011 Broncos, whose option-flavoured season was the wave's warm-up act), and the four wave teams accounted for about 65 percent of the league's designed quarterback runs in 2012. The rest of the league mostly watched. The league-wide rate roughly doubled from its 2008-2010 low to a peak in 2013, but it peaked at about one designed quarterback run per 100 plays, less than one a game. The read-option wave was a revolution in four offenses and a rumour in the other twenty-eight.²⁰

And it was effective while it lasted: designed quarterback runs in 2012 averaged +0.10 EPA per play, against -0.07 for running back runs in the same situations. So why did it recede? Not because it stopped working on paper. Four things happened, and they are a good lesson in how the NFL absorbs a new idea.

1. **The quarterbacks got hurt, or the teams got scared they would.** Griffin, already playing on a sprained knee, tore two ligaments in it in the playoff loss to Seattle in January 2013 and was never the same player. A franchise quarterback is the most valuable asset in football; a run game that exposes him to linebackers ten times a game is a risk most owners and coaches would rather not take.
2. **Defenses went to school.** The 2013 offseason was a crash course. Green Bay's coaches, run over by Kaepernick in the playoffs, spent time with Texas A&M's staff studying how college defenses played the read.²¹ Defenses learned to give the quarterback a confusing picture: the scrape exchange, in which the read defender crashes on the back and a linebacker loops behind him to take the quarterback, and "slow-playing" the read, keeping the unblocked defender in no-man's land until the quarterback had to commit.

²⁰This book's calculation from nflverse play-by-play, regular seasons 2006-2025. Designed QB run: a run play whose ball carrier had 50 or more pass attempts that season, not flagged as a scramble or an aborted snap, on 3 or more yards to go (kneel-downs are a separate play type). League rate per 100 offensive plays: 0.46-0.51 (2008-2010), 0.82 (2012), 1.00 (2013), 0.79-0.87 (2015-2017), 1.39 (2020 and 2022). Highest team-seasons from 2006 through 2012: Denver 2011 (Tim Tebow, 6.95), Atlanta 2006 (Michael Vick, 6.14), Washington 2012 (6.11), Carolina 2012 (5.84); no other team reached 5 in 2006-2011. (Newton's later Carolina teams went higher: 6.76 in 2017.) Lamar Jackson's Ravens: 8.18 (2018), 9.45 (2019), 8.90 (2020). Washington, Carolina, San Francisco and Seattle ran 172 of the league's 266 designed QB runs in 2012 (65 percent). Read-option *gives* to the running back cannot be identified in play-by-play before charting data, so this counts keeps only. Reproduce with the code behind the figure.

²¹NFL.com, "Mike McCarthy leans on Texas A&M for read-option tips" (2013): <https://nfl.com/news/mike-mccarthy-leans-on-texas-a-m-for-read-option-tips-0ap1000000143890> (announced at the 2013 Combine; Dom Capers's defensive staff to work with Kevin Sumlin's staff on defending the read option).

3. **The rules said so.** The league had quietly told its officials in 2012 that a read-option quarterback counts as a runner; before the 2013 season it spelled it out to every team in an officiating video: a quarterback carrying out a read-option fake may be hit like a runner until he is clearly out of the play.²² Every keep, and every fake, became a chance to hit the quarterback.
4. **The quarterbacks evolved.** Wilson became a scrambler who threw from the pocket. Kaepernick's offense leaned on his arm. Newton was the exception who proved the rule: Carolina kept him running, and in 2015 he was the league MVP and took the Panthers to Super Bowl 50, where Denver's hybrid edges were waiting.

What survived was the read itself. The zone read became a change-up most offenses keep in the playbook, the RPO moved the same "read one defender" idea into the passing game, and from 2018 Lamar Jackson's Ravens built an offense around designed quarterback runs at a scale the 2012 teams never approached. The wave receded; the tide came in later.

Chip Kelly's 2013 Eagles: the college offense, whole

One coach tried to bring the entire college package at once. Chip Kelly came to Philadelphia in 2013 from Oregon, where his spread-option offense had snapped the ball faster than almost anyone in college football. His first Eagles team went 10-6, won the NFC East and led the league in rushing, and its quarterback, Nick Foles, threw 27 touchdown passes and two interceptions.²³ The full story, with charts of how fast it went and what that cost Kelly's defense, is the Chip Kelly film room in In-Game Adjustments and Tempo.

In the arc of this chapter, Kelly's Eagles matter because they combined three of its threads in one offense. **Tempo**, which froze the defense's personnel the way the Patriots' no-huddle did, but on almost every snap. **The read**, from college. And the **packaged play**: an inside zone run with a bubble screen or a quick pass attached, which the quarterback chose after the snap by reading one defender, the play that grew up into today's RPO (RPOs). Kelly's teams faded fast (he was fired before the end of his third season), and the bill for the tempo landed on his own defense, which faced more plays than any other in two of his three seasons. But the ideas didn't fade. When the Eagles won Super Bowl LII four years later with Doug Pederson, RPOs were all over their game plan, and by the 2020s packaged plays were in every NFL playbook.

The first analysts

The last thread of the era didn't happen on the field at all. While coaches were arguing about personnel, a few outsiders were asking a different question: *are coaches making the right decisions at all?*

David Romer, 2006. Romer, an economist at Berkeley, wanted to test a basic idea in economics, that competition forces organizations to make the best decisions. He picked NFL fourth downs as his test case, because they were measurable and the stakes were high. Using play-by-play data from the 1998 to 2000 seasons, he estimated the value of going for it, punting and kicking a field goal from every spot on the field, and found that teams kicked far too often: the choices that maximized a team's chance of winning were systematically different from the ones coaches made. He published it in the *Journal of Political Economy* in 2006 under the title "Do Firms Maximize? Evidence from Professional Football."²⁴ Coaches mostly ignored it. One of the few who had actually studied it, even in its 2002 working-paper form, was Belichick.

The bloggers and the models. At about the same time, analysis moved onto the internet. Aaron Schatz's Football Outsiders, launched in 2003, built DVOA, a play-by-play efficiency rating adjusted for situation and opponent (Expected Points and Success Rate places it among the metrics). Brian Burke, a former Navy F/A-18 pilot, started *Advanced NFL Stats* in 2007 and published

²²NESN, "Jim Harbaugh rips NFL's new read-option rules as 'flawed,' 'biased'" (Sept. 2013): <https://nesn.com/2013/09/jim-harbaugh-rips-nfls-new-read-option-rules-as-flawed-biased> (Dean Blandino, NFL vice president of officiating, in a league video on the rule: "The quarterback can be hit like a runner until he's clearly out of the play"); see also NFL.com, "Niners' Jim Harbaugh, NFL exec discuss QB hit rules" (Sept. 2013): <https://www.nfl.com/news-migrated-v2/niners-jim-harbaugh-nfl-exec-discuss-qb-hit-rules-0ap1000000238827>. On the 2012 date: Associated Press, "NFL says Terrell Suggs' hit on Sam Bradford was legal" (Aug. 24, 2015): <https://abc7.com/post/nfl-says-terrell-suggs-hit-on-sam-bradford-was-legal/954413/> (Blandino: "We clarified it in 2012 with the proliferation of these read-option schemes").

²³"2013 Philadelphia Eagles season," Wikipedia: https://en.wikipedia.org/wiki/2013_Philadelphia_Eagles_season (10-6, NFC East champions, first in the league in rushing; Foles 27 touchdown passes, 2 interceptions). Plays faced by Kelly's defenses: KTVU, "Chip Kelly's defensive woes continue in SF" (2016), citing Sportingcharts: <https://ktvu.com/sports/chip-kellys-defensive-woes-continue-in-sf> (most in the league in 2013, 1,150, and 2015, 1,148; second in 2014, 1,113). How much of that was the offense's tempo rather than the defense's own failures was disputed at the time: Philadelphia Magazine, "So far, 'plays run' has been a losing battle": <https://www.phillymag.com/?p=3048213>. Kelly was fired on December 29, 2015, with one game left in his third season.

²⁴David Romer, "Do Firms Maximize? Evidence from Professional Football," *Journal of Political Economy* 114, no. 2 (April 2006): 340-365, https://eml.berkeley.edu/~dromer/papers/JPE_April06.pdf (play-by-play from the 1998-2000 seasons; teams' fourth-down choices depart systematically from those that maximize the chance of winning). Pages and DOI (10.1086/501171): RePEc, <https://ideas.repec.org/a/ucp/jpolec/v114y2006i2p340-365.html>. An earlier version circulated in June 2002 as NBER Working Paper 9024, "It's Fourth Down and What Does the Bellman Equation Say? A Dynamic Programming Analysis of Football Strategy": <https://www.nber.org/papers/w9024>. On Belichick: Greg Garber, ESPN.com, "Fourth-down analysis met with skepticism" (Oct. 31, 2002): https://www.espn.com/nfl/columns/garber_greg/1453717.html (Belichick "is one of several NFL head coaches who has actually studied Romer's working paper").

models of **expected points** and **win probability** in the open, the same ideas this book has used since Downs, Distance, and Field Position. Those open models are the forerunners of the nflverse models behind every chart in this book.²⁵

i Film room: Patriots at Colts, November 15, 2009 — fourth-and-2

The decision that made analytics a front-page argument. New England led Indianapolis 34-28 with about two minutes left and faced fourth-and-2 at its own 28-yard line. The conventional call was obvious: punt, and make Manning drive the length of the field. Belichick went for it. Brady's pass to Kevin Faulk was ruled short of the line to gain, the Colts took over inside the Patriots' 30, scored, and won 35-34. The next morning almost every columnist in the country called it a blunder.

Burke's win-probability model disagreed. By his estimate, going for it gave the Patriots about a 79 percent chance to win and punting about 70 percent, because a successful conversion ended the game and Manning, given the ball even 70 yards away, scored often enough to make the punt worse than it looked.²⁶ (Estimates like these have wide error bars, and some analysts put the two options closer together. Almost none put the punt clearly ahead.)

What to notice: not whether the play worked. It didn't. Notice the shape of the argument: a decision judged by its outcome in the newspapers and by its *probabilities* in the models. That gap between outcome and decision is the whole field of football analytics, and it took the league another decade to start acting on it. Win Probability and Fourth Downs builds the model, and The Modern Era shows what happened when coaches finally listened.

The pattern: every answer becomes the next problem

Step back and the seventeen seasons have a rhythm you can use on any Sunday.

The punch	The counterpunch	What it opened up
Tampa 2: fast, simple, everything in front	Manning: no-huddle, play-action, a receiver-speed tight end up the seam	Defenses needed a faster middle defender, or a different coverage
Physical coverage downfield	2004 illegal-contact emphasis	A passing boom, and the 2007 spread
The 2007 Patriots' spread	The Giants' four-man rush	A premium on edge rushers
12 personnel with a move tight end	Bigger safeties; Seattle's one-coverage Cover 3	Offenses built on tempo and college ideas
The 3-4's hybrid edges	Quick passing, then the read option, which turns an edge rusher's aggression against him	The read became the RPO
The read-option wave	Scrape exchange, hitting the quarterback, college study	Designed QB runs return, bigger, after 2018

None of these answers was permanent, and none of the punches disappeared. Cover 2 is still called on third downs; the move tight end is in every offense; the hybrid edge is now the most valuable defender on most teams; the read lives on inside the RPO. History in the NFL rarely ends. It gets folded into the next scheme.

²⁵ESPN Front Row, "Former Navy pilot Brian Burke joins ESPN as senior analytics specialist" (June 2015): <https://www.espnfrontrow.com/2015/06/former-navy-pilot-brian-burke-joins-espn-as-senior-analytics-specialist/> and ESPN Press Room bio, <https://espnpressroom.com/us/bios/brian-burke/> (Naval Academy graduate; much of 15 years as an F/A-18 carrier pilot; launched the site in 2007, originally Advanced NFL Stats, later Advanced Football Analytics); "Football Outsiders," Wikipedia: https://en.wikipedia.org/wiki/Football_Outsiders (founded by Aaron Schatz in 2003; DVOA). The lineage from these public models to nflscrapR and nflfastR, whose expected-points model nflverse uses, is in Expected Points and Success Rate.

²⁶Boston.com, "Belichick has math on his side" (Nov. 16, 2009): https://www.boston.com/sports/new-england-patriots/2009/11/16/belichick_has_math_on_his_side/ (Burke's estimate: about 79 percent to win by going for it, about 70 percent by punting); game details (Patriots led 34-28, fourth-and-2 at their own 28 with about two minutes left; Faulk ruled down short of the 30; Colts won 35-34): "2009 New England Patriots season," Wikipedia: https://en.wikipedia.org/wiki/2009_New_England_Patriots_season.

💡 Watch for it on Sunday: the personnel dilemma

Pick one team whose tight end is a real receiver, and watch every snap it uses 12 personnel (one back, two tight ends; the broadcast graphic or the two tight ends running on will tell you).

- **Before the snap: what did the defense send out?** Count the defensive backs. Four means base (three linebackers); five means nickel; a big safety playing where a linebacker would is the modern in-between answer, the big nickel.
- **Then: where does the tight end line up?** Next to the tackle, in the slot, out wide, or in the backfield? If he moves after the defense has chosen, the offense is hunting a matchup.
- **If the defense is in base**, watch for the tight end split out against a linebacker. That is the pass the offense wants.
- **If the defense is in nickel**, watch for both tight ends tight to the formation and a run at the nickel back's side.
- **Is the offense in no-huddle?** Then the defense is stuck with its last answer. Watch whether it checks to a different coverage instead (more man, a safety on the tight end).
- **After the snap**, ask: who ended up covering the tight end, and was that player chosen for him or stuck with him?

Predict the play

💡 Predict the play 1: no huddle, base defense

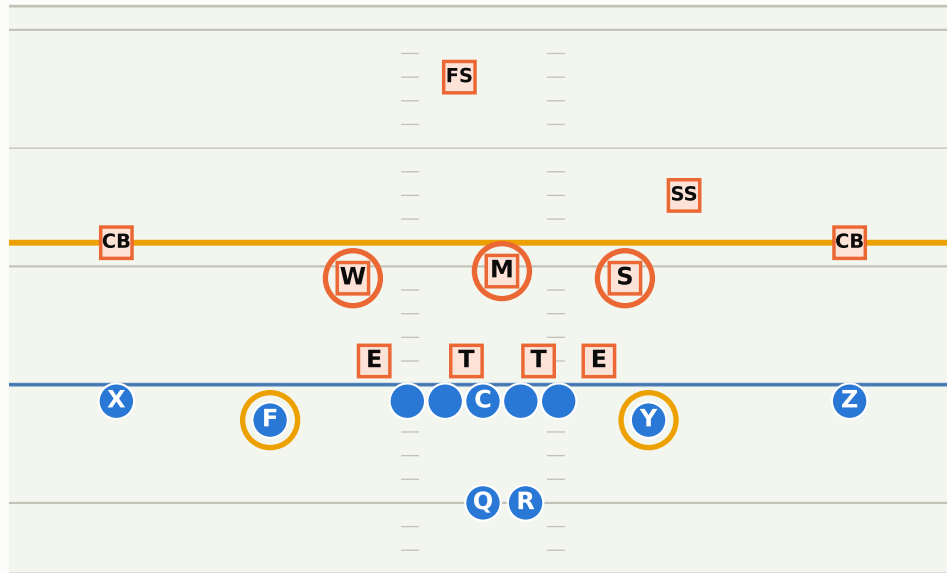


Figure 7: 2011-style Patriots, second-and-6, no huddle. The offense has 12 personnel on the field; the defense, which matched the previous play's personnel, is still in its base 4-3 with a single deep safety. The offense has just lined up like this: Hernandez (F) in the left slot, Gronkowski (Y) detached a few yards outside the right tackle, the running back beside the quarterback. The two tight ends are ringed in gold, the three linebackers in orange.

Who is the defense's weakest matchup, what would you throw, and what could the defense have done about it?

Answer 1 is at the end of the chapter. Try it first.

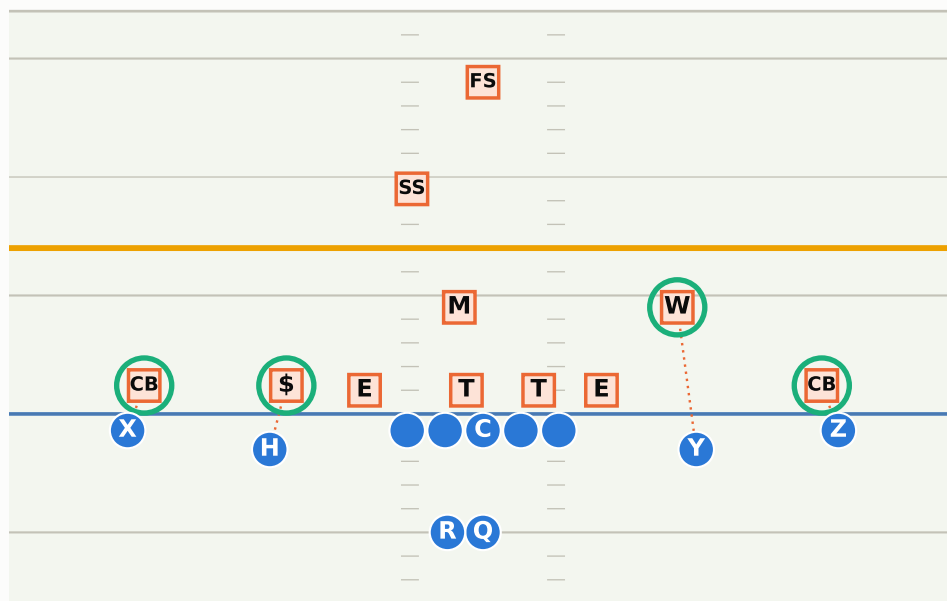


Figure 8: Third-and-7, early in the 2004 season. The offense has three wide receivers and a tight end. The defense shows man coverage across the board: both cornerbacks and the nickel back pressed on the receivers (a yard off, inside shoulder), a linebacker on the tight end, one safety deep in the middle. This is how a physical secondary played the Colts in January 2004. The man defenders are ringed.

This coverage worked beautifully against the Colts in the 2003 playoffs. Why is it riskier in 2004, and which routes become more dangerous against it?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: the unblocked edge

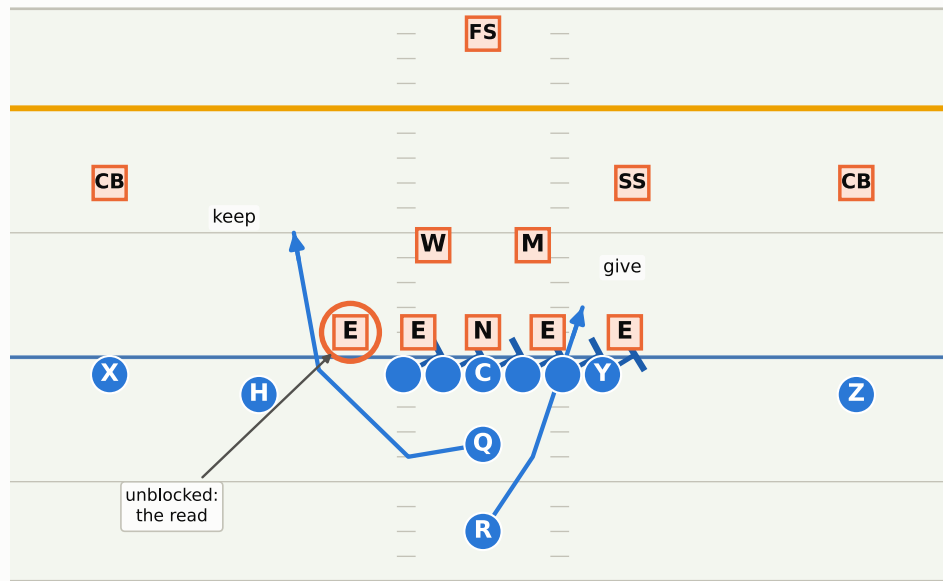


Figure 9: 2012, first-and-10. The offense is in the pistol (the quarterback about 4 yards behind the center, the back behind him) and its line will zone-block to the right at the snap, leaving the defense's right edge player, a standing 3-4 outside linebacker (ringed), unblocked on purpose. The two possible ball carriers' paths are drawn: the back's give track to the right, and the quarterback's keep around the left end.

You are the defensive coordinator in the 2013 offseason. What do you tell the ringed edge player to do, who helps him, and what has the league just told you about the quarterback?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Tampa 2 dominated by being simple and fast, and lost to an offense that was simpler and faster.** Manning's Colts used no-huddle, line-of-scrimmage checks, play-action that made the Mike step up, and a receiver-speed tight end up the seam to put the throw where Tampa 2 was thinnest.
- **The 2004 illegal-contact emphasis enforced an old rule, and it tilted the field.** Illegal-contact fouls more than doubled, league passer rating jumped four points in a season, and passing efficiency kept rising through 2016 while running efficiency stood still.
- **The Patriots won by changing shape:** an Erhardt-Perkins language that let them rebuild every year, the 2007 spread, and the 2010-2012 two-tight-end offense, whose **move tight end** made a defense choose base or nickel before it knew where he'd line up. The counterpunches were a four-man rush (the 2007 Giants) and safeties who could play like linebackers (Seattle).
- **The 3-4 came back because of the hybrid edge,** a player too light for a 4-3 end and too explosive for linebacker, who could rush or drop and keep a passing offense guessing. By the 2010s "EDGE" was a position of its own.
- **The 2012 read-option wave was real in four offenses and small in the league.** It was efficient, but it receded because quarterbacks got hurt, defenses learned the scrape exchange and the slow-play, and the league said read quarterbacks could be hit. The read survived inside the RPO, and designed quarterback runs came back bigger after 2018.
- **Analytics started outside the building:** Romer's fourth-down paper, Football Outsiders and Brian Burke's open models. Belichick's 2009 fourth-and-2 made the argument public, a decade before coaches acted on it.

Connections

Builds on: The College Laboratory (where the spread, the spread option and the RPO came from), Cover 2 and Tampa 2 (the coverage this chapter's first section attacks), Personnel Groupings (the substitution rule and the two-tight-end dilemma), Option Football (the zone read and the scrape exchange), and Odd and Hybrid Fronts (the stand-up edge).

Sets up: The Modern Era, where the two-high counterrevolution answers the offenses this chapter built, and where the hybrid edge becomes the premium defender and the quarterback run comes back; Belichick and Game-Plan Football, the full Patriots case study; The Cover 3 Family for Seattle's defense; In-Game Adjustments and Tempo for the no-huddle and Kelly's Eagles; RPOs, where the packaged play grew up; Contact Rules and Penalties as Strategy for today's contact rules; and Win Probability and Fourth Downs, which turns the 2009 argument into a model you can run. The rules timeline in Appendix E puts the era's rule changes in one list.

If you want to go deeper: Tim Layden, *Blood, Sweat and Chalk* (2010), for the coaches behind Tampa 2, the zone blitz and the 3-4 in their own words; Chris B. Brown, *The Art of Smart Football* (2015) and *The Essential Smart Football* (2012), for essays on Manning's offense, the Patriots and the spread's arrival in the NFL; Michael Holley, *Patriot Reign* (2004) and *War Room* (2011), on how Belichick's staff thought; Pete Carroll, *Win Forever* (2010), for Seattle's philosophy; David Romer's paper itself, which is readable and short; and Brian Burke's archived *Advanced Football Analytics* essays, the clearest early explanations of expected points and win probability.

Answers to Predict the play

Answer 1.

The Will on Hernandez. With three linebackers and only two corners and two safeties, someone slow has to cover someone fast. The corners take X and Z, the strong safety is the natural choice for Gronkowski, and that leaves a linebacker, probably the Will, walking out to Hernandez in the slot with nobody near him to help. A quick option route or a slant by F against a linebacker in space is the throw; if the safety slides to help on F, Gronkowski is on a linebacker instead, and the seam is his.

The defense's options were all bad, which was the point. It couldn't substitute (no huddle). It could check to a zone and give up easy completions underneath, or play man with the Sam on Gronkowski and the Will on Hernandez and pray. The real answers came from roster-building, not play-calling: safeties who can tackle like linebackers and linebackers who can cover like safeties, so the defense doesn't have to choose personnel at all. That is exactly what Seattle built, and why the big nickel is everywhere now.

Answer 2.

Because the hands have to come off at 5 yards. Press man works best when the defender can stay physical with the receiver all the way through his route: a jam at the line, a bump at 7 yards, a hand on the hip at the break. In 2003 officials let a lot of that go. From 2004, with illegal contact a point of emphasis, any contact past 5 yards before the throw is a flag: 5 yards and an automatic first down, which on third-and-7 is as good as a completion. The corner gets one shot, at the line. If he misses it, he has to run with the receiver clean.

The routes that win are the ones that break after 5 yards and away from the defender's leverage. With the corners on the receivers' inside shoulders, out-breaking routes (outs, comebacks, corner routes) at 8 to 15 yards are hard to cover clean. Crossing routes make man defenders run through traffic. And the linebacker on the tight end, Will, is the weakest matchup on the field, the move tight end's lesson again. Defenses responded by playing more zone, more off coverage, and more pressure (the quarterback has to throw before the clean route can develop), which is part of why the zone-blitzing 3-4 spread in the same years.

Answer 3.

Don't let the quarterback read you. If the edge simply chases the back, the quarterback keeps around him; if he sits, the quarterback hands off and the back runs into a box one defender short. The 2013 answers, from Option Football, were all ways of giving the quarterback the wrong picture:

- **Scrape exchange.** The edge crashes hard down the line at the back, which looks like "keep" to the quarterback, and an inside linebacker loops behind him to the spot where the keep goes. The quarterback keeps, into a linebacker he didn't count.
- **Slow-play.** The edge shuffles down the line, square, staying between the two ball carriers as long as he can, so the quarterback has to decide before the edge does.
- **Hit him.** The league's 2013 guidance said a quarterback who carries out a read fake can be hit like a runner until he's clearly out of the play. So the edge is told: if in doubt, hit the quarterback. Every keep costs the offense a hit on its most valuable player, and many offenses decided that price was too high.

And the long-term answer is in the picture: the defender you most want here is a hybrid edge, big enough to take on a lead blocker and fast enough to chase a quarterback. Defenses spent the next decade collecting them.